

CHENGHAO GUO

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RESEARCH INTERESTS

My research lies at the intersection of **statistics and computer science**. I am especially interested in understanding statistical problems from a computational view. Specifically, I work on **average-case and smoothed complexity** of signal recovery and optimization problems.

Average-Case Complexity · Smoothed Complexity · Probability in High Dimension · Information Theory · Random Graph Theory · Markov Chains

EDUCATION

Massachusetts Institute of Technology <i>Ph.D. in Electrical Engineering and Computer Science</i> Advisors: Guy Bresler and Yury Polyanskiy	<i>Sep 2022 - May 2025</i>
Massachusetts Institute of Technology <i>M.Eng. in Electrical Engineering and Computer Science</i> Advisors: Guy Bresler and Yury Polyanskiy	<i>Sep 2020 - May 2022</i>
Institute for Interdisciplinary Information Sciences (Yao Class), Tsinghua University <i>B.Eng. in Computer Science</i>	<i>Sep 2016 - June 2020</i>

PUBLICATIONS AND PREPRINTS

Authors are in *alphabetic* order unless otherwise specified

Partial and Exact Recovery of a Random Hypergraph from its Graph Projection [\[arXiv\]](#)

Guy Bresler, **Chenghao Guo**, Yury Polyanskiy, Andrew Yao
Conference on Learning Theory, COLT 2025

Thresholds for Reconstruction of Random Hypergraphs From Graph Projections [\[arXiv\]](#)

Guy Bresler, **Chenghao Guo**, Yury Polyanskiy
Conference on Learning Theory, COLT 2024

Smoothed Complexity of SWAP in Local Graph Partitioning [\[arXiv\]](#)

Xi Chen, **Chenghao Guo**, Emmanouil V Vlatakis-Gkaragkounis, Mihalis Yannakakis
Symposium on Discrete Algorithms, SODA 2024

Algorithmic Decorrelation and Planted Clique in Dependent Random Graphs: The Case of Extra Triangles [\[arxiv\]](#)

Guy Bresler, **Chenghao Guo**, Yury Polyanskiy
Symposium on Foundations of Computer Science, FOCS 2023

Linear Programs with Polynomial Coefficients and Applications to 1D Cellular Automata [\[arxiv\]](#)

Guy Bresler, **Chenghao Guo**, Yury Polyanskiy
preprint, 2022

Generalizing Complex Hypotheses on Product Distributions: Auctions, Prophet Inequalities, and Pandora's Problem [\[arXiv\]](#)

Chenghao Guo, Zhiyi Huang, Zhihao Gavin Tang, Xinzhi Zhang
Conference on Learning Theory, COLT 2021

Smoothed Complexity of Local Max-Cut and Binary Max-CSP [\[arXiv\]](#)

Xi Chen, **Chenghao Guo**, Emmanouil V Vlatakis-Gkaragkounis, Mihalis Yannakakis, Xinzhi Zhang
Symposium on the Theory of Computing, STOC 2020

Settling the Sample Complexity of Single-parameter Revenue Maximization [\[arXiv\]](#)

Chenghao Guo, Zhiyi Huang, Xinzhi Zhang

Symposium on the Theory of Computing, STOC 2019

Also appear as: Sample Complexity of Single-parameter Revenue Maximization (*SIGecom Exchanges Vol 17.2*)

Decomposition of a Symmetric Multipartite Observable [\[arXiv\]](#)

You Zhou, Chenghao Guo, Xiongfeng Ma (joint first author with You Zhou)

Phys. Rev. A, May 2019

SELECTED HONORS

MathWorks Fellow	2024
Silver Medal of Yao Award	2019
Toyota Scholarship	2019
Star of Baidu Scholarship	2017, 2018
Gold Medal in Chinese Physical Olympiad	2015

RESEARCH EXPERIENCES

Computational Statistics and Information Theory, MIT

June 2020 - present

- Laboratory for Information and Decision Systems (LIDS)
- Broadcasting of information on grids.
- Average-case complexity of signal recovery on graphical structures.

Machine Learning, Google

August 2023 - present

- Student researcher
- The theoretical foundations of distillation in learning tasks.

Algorithm and Complexity, Columbia University

Feb 2019 - present

- Smoothed analysis of local max-cut and other optimization problems.

Algorithmic Game Theory, the University of Hong Kong

July 2018 - June 2020

- Optimal sample complexity of mechanism design.

Quantum Information, Tsinghua University

Nov 2016 - Aug 2018

- Protocols for state measurement of quantum symmetric states.

TEACHING EXPERIENCE

Teaching Assistant at MIT for

- **Machine Learning** (6.7900) in Fall 2025
 - Introductory machine learning course for graduate students, ~ 250 students.
 - Office hours, organizing homework and exams, coordinating graders.
- **Algorithms for Inference** (6.7810) in Fall 2023, review rating: **6.8/7.0**
 - Graduate course on algorithms and theoretical foundations in inference tasks.
 - Recitations, office hours, developing problem sets, developing exam questions, organizing exams.

REVIEWER FOR

Symposium on the Theory of Computing (STOC)	2023, 2024, 2025
Conference on Learning Theory (COLT)	2024
Information and Inference: A Journal of the IMA	2024
Algorithmica	2024
Innovations in Theoretical Computer Science Conference (ITCS)	2026
IEEE International Symposium on Information Theory (ISIT)	2023

REFERENCES

Guy Bresler

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